

Ashwot Acharya

Computational Mathematics student

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Profile

Computational Mathematics student building ML systems from first principles — spectral learning, kernel-level anomaly detection, and graph learning theory. Currently researching connections between Weighted Finite Automata (WFA) expressiveness and Graph Neural Networks (GNNs) via the Weisfeiler-Leman hierarchy. Experienced implementing neural networks, RNNs, and formal language models without external ML libraries.

Research

WFA-GNN Research, Ongoing Research 04/2026 – Present
• Investigating formal connections between Weighted Finite Automata and Graph Neural Networks.
• Implementing spectral learning pipeline: Hankel matrix construction to SVD to WFA parameter recovery;
• building RNN from scratch to validate WFA extraction on learned sequence functions.

Projects

Multi-Agent Intrusion Detection System 2026 – 2026
• Implementation of 5-agent intrusion detection system analyzing bulk network flow metadata , TLS 1.3 traffic
• Implemented of context aware BiLSTM model for behavioral anomaly detection
• Built an Isolation Forest model for flow-level anomaly scoring
• Trained models using 5-fold cross-validation to ensure robust generalization across attack scenarios
• Performed exploratory data analysis and feature engineering to clean and structure raw network traffic data for model training

Kernel-Level Intrusion Detection System (CFG-IDS) 2026 – 2026
• Built a kernel-level IDS using PCFG scoring and SEQUITUR grammar induction over live syscall sequences captured via perf/strace.
• Implemented SEQUITUR from scratch, CNF conversion, PCFG learning with Laplace-smoothed NGram fallback, and a cyclic monitor supporting both batch and sliding-window modes.
• Validated against real attack scenarios: RCE, path traversal, SSRF, malware injection, and reverse shells on a vulnerable Node.js target.

Sudoku Solver via SAT (CDCL) 2025 – 2025
• Modeled Sudoku as a SAT problem using optimized CNF encoding (Lynce & Ouaknine, 2006), reducing clause count by up to 56%.
• Implemented core CDCL components: unit propagation, conflict analysis, clause learning, and non-chronological backtracking.
• Benchmarked on 9x9, 16x16, 32x32 grids and low-clue instances (7-clue Sudoku); compared against backpropagation baselines.

Proof Assistants and Formalization in Agda 2025 – 2025
• Formalized mathematical proofs in Agda, including algebraic properties and logical propositions in a dependent type-theoretic setting.
• Conducted comparative analysis of Agda, Coq (Rocq), and Lean — focusing on kernel design, type-checking algorithms, and proof-term extraction.

Skills

ML / Math NumPy, PyTorch, Pandas, scikit-learn; SVD, spectral methods, PCFG, CNF/SAT	Web / Tools FastAPI, React, WebSockets, Git, LaTeX, Beamer
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Education

BSc in Computational Mathematics, Kathmandu University, Dhulikhel 2023 – Present | Dhulikhel, Nepal
• Coursework: Matrix Theory, Complex Analysis, Formal Language Theory, Operations Research, Computer Architecture
• Research Focus: Graph Theory, Optimization, Graph Neural Networks, Spectral Learning

Experience

Data Annotation Intern, Togglecorp 02/2023 – 02/2023 | Lalitpur, Nepal
• Performed data annotation for computer vision tasks, supporting model training pipelines.

Web Development Intern, Ezone International 12/2022 – 01/2023 | Lalitpur, Nepal
• Improved legacy UI components and contributed to frontend redesign; integrated updated features with the backend team.